LESSON 01 / 03

## A normalized pencil of circles

Write $S_i = x^2 + y^2 + D_i x + E_i y + F_i$ for $i = 1, 2$, normalizing each circle to the same $x^2 + y^2$ coefficient. Then

$$S_\lambda = (1 - \lambda)S_1 + \lambda S_2 = 0$$

is a family of circles. Every member has the same radical axis $S_1 - S_2 = 0$, and if the original circles intersect in two real points, every member passes through both. If they do not meet in real points, the radical axis is still an equal-power line but is not a real common chord.

**WORKED EXAMPLE**

***A pencil through two fixed points***

The family $x^2 + y^2 + \lambda(x - y) - 2 = 0$ passes through $(1, 1)$ and $(-1, -1)$ for every real $\lambda$. Subtracting two members gives $(\lambda_1 - \lambda_2)(x - y) = 0$, so their radical axis is $x - y = 0$.

## Radical centre, when finite

For three circles, solve two radical-axis equations. If the axes are nonparallel, their intersection is the unique finite radical centre, and equal-power transitivity makes powers to all three circles equal. If axes are parallel or coincide, do not force a finite point.

**COMMON TRAP**

***Concentric degeneracy from the source***

For $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, equal powers would require $-1 = -4$, impossible: these concentric unequal circles have no finite radical axis. A third circle cannot create a finite point equidistant in power from both. Replace one circle with a nonconcentric one before asking for a radical centre.

LESSON 02 / 03

## Inversion in a circle

Inversion of radius $\rho$ about centre $O$ sends $P \neq O$ to $P'$ on ray $OP$ so that $OP \cdot OP' = \rho^2$. With $O$ at the origin,

$$P' = \frac{\rho^2 P}{|P|^2}.$$

It is involutive and reverses orientation while preserving angles in magnitude. The centre has no finite image.

SHORT DERIVATION

**A line not through the inversion centre becomes a circle**

For unit inversion, write $x = X/(X^2 + Y^2)$ and $y = Y/(X^2 + Y^2)$. The line $x = a \neq 0$ becomes $X = a(X^2 + Y^2)$, or $(X - 1/(2a))^2 + Y^2 = 1/(4a^2)$: a circle through the origin. A line through the origin stays a line; conversely, a circle through the origin maps to a line not through it.

## Distance ratio is a circle, with a line as the limiting case

For distinct $A, B$, the locus $PA/PB = k$ is the perpendicular bisector when $k = 1$ and a circle when $k \neq 1$. The centre lies on $AB$ and the circle degenerates to $A$ at $k = 0$. This is the Apollonius locus; a derivation and coordinate-safe formula are in Chapter 12.

COMMON TRAP

**Check for the exceptional ratio**

Do not divide by $1 - k^2$ before splitting off $k = 1$. For equal distances the locus is a line, not a circle of "infinite radius" in an ordinary coordinate calculation.

LESSON 03 / 03

## Simson line: an exact hypothesis

Let $ABC$ be a nondegenerate triangle and $P$ any point. The feet of perpendiculars from $P$ to the three **lines** $BC, CA, AB$ are collinear if and only if $P$ lies on the circumcircle of $ABC$. The word "lines" matters: a foot may fall on an extension of a side.

SHORT DERIVATION

**The cyclic quadrilaterals behind Simson**

Call the feet $D \in BC$, $E \in CA$, $F \in AB$. The right angles show $P, D, C, E$ and $P, D, B, F$ are cyclic. Directed angles modulo $180^\circ$ give

$$\angle EDF = \angle EDP + \angle PDF.$$

The two right-angle circles give $\angle EDP = \angle ECP$ and $\angle PDF = \angle PBF$. Since $E, C, A$ and $F, B, A$ lie on the same side-lines,

$$\angle ECP = \angle ACP, \qquad \angle PBF = \angle PBA.$$

If $A, B, C, P$ are concyclic, $\angle ACP = \angle ABP$, while $\angle PBA = -\angle ABP$ as directed angles, so $\angle EDF = 0$ and the feet are collinear. Reversing the same equalities proves the converse in the nondegenerate case. Numerical collinearity alone is evidence, not a proof.

## Ptolemy: order the vertices around the circle

For a convex cyclic quadrilateral $A, B, C, D$ in circular order,

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

SHORT DERIVATION

**A similarity proof**

Choose $E$ on diagonal $AC$ so that $\angle ABE = \angle CBD$. Since $A, B, C, D$ are cyclic and $E$ lies on $AC$, triangles $ABE$ and $CBD$ are similar, giving $AE \cdot BD = AB \cdot CD$. The remaining triangles $BCE$ and $BDA$ are similar, giving $CE \cdot BD = BC \cdot DA$. Add and use $AE + CE = AC$ to obtain Ptolemy.

WORKED EXAMPLE

**Pythagoras is the rectangle case**

For a rectangle with adjacent sides $a, b$ and both diagonals $c$, Ptolemy gives $c^2 = a^2 + b^2$. The rectangle condition matters: a general cyclic quadrilateral with one diagonal a diameter does not by itself make Ptolemy reduce directly to that equation.

# Simson Line and Ptolemy (cont.)

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

COMMON TRAP

**Source-paper consistency correction**

A cyclic quadrilateral with successive sides $3, 4, 5, 6$ and a diagonal of length $7$ cannot exist: that diagonal and the adjacent sides $3, 4$ would form a degenerate triangle ($3 + 4 = 7$). Ptolemy's numerical equation alone is not an existence proof for arbitrary proposed side/diagonal data.